

Week 4: User & Group Management (Part 2)

Date: March 30, 2026

Study Time: 5-8 hours total

GOAL

Implement secure sudo policies across all devices

STUDY TASKS (1-2 hours)

Reading

- ☐ "How Linux Works" by Brian Ward
 - Chapter 7: sudo configuration section

Video Learning

- ☐ LearnLinuxTV - Search: "sudo configuration"
 - <https://youtube.com/@LearnLinuxTV>
- ☐ Professor Messer - Search: "Linux+ sudo"
 - <https://youtube.com/@professormesser>

Concepts to Understand

- ☐ Difference between `su` and `sudo`
 - ☐ Principle of least privilege
 - ☐ `/etc/sudoers` file structure
 - ☐ sudo logging and auditing
 - ☐ Command restrictions
 - ☐ Password vs NOPASSWD
-

LAB WORK (3-4 hours)

Understanding sudo

- ☐ Check current sudo access

```
bash
```

```
sudo -l  
# Lists what you can run with sudo
```

☐ View sudoers file safely

```
bash
```

```
sudo cat /etc/sudoers  
# NEVER edit directly with vim/nano!
```

Edit Sudoers Safely

☐ Use visudo

```
bash
```

```
sudo visudo  
# Opens sudoers with syntax checking  
# Prevents saving broken configs
```

Grant Full sudo Access

☐ Give user all sudo privileges

```
bash
```

```
sudo visudo  
  
# Add this line:  
tier1admin ALL=(ALL:ALL) ALL  
  
# Format: USER HOST=(RUNAS) COMMANDS  
# tier1admin = username  
# ALL = from any host  
# (ALL:ALL) = can run as any user:group  
# ALL = can run all commands
```

☐ Test sudo access

bash

```
su - tier1admin
sudo whoami # Should return 'root'
sudo systemctl status sshd
exit
```

Create Command-Specific Rules

☐ Allow specific commands only

bash

```
sudo visudo

# Allow only systemctl
monitoring ALL=(ALL) /usr/bin/systemctl

# Allow multiple specific commands
developer ALL=(ALL) /usr/bin/systemctl, /usr/bin/journalctl

# Allow without password (use cautiously!)
backup_user ALL=(ALL) NOPASSWD: /usr/bin/rsync
```

☐ Test restrictions

bash

```
su - monitoring
sudo systemctl status sshd # Should work
sudo apt update            # Should fail
exit
```

Group-Based sudo

☐ Grant sudo to entire group

bash

```
sudo visudo
```

```
# Add this line:
```

```
%admin ALL=(ALL:ALL) ALL
```

```
# % means it's a group, not a user
```

```
# Now anyone in 'admin' group has sudo
```

Enable sudo Logging

☐ On Fedora (systemd journal)

bash

```
# Logs go to journalctl automatically
```

```
sudo journalctl -u sudo
```

```
# Or filter by user
```

```
sudo journalctl | grep sudo
```

☐ Create dedicated sudo log

bash

```
sudo visudo
```

```
# Add to Defaults section:
```

```
Defaults logfile="/var/log/sudo.log"
```

```
Defaults log_year, log_host, loglinelen=0
```

```
# Create log file
```

```
sudo touch /var/log/sudo.log
```

```
sudo chmod 640 /var/log/sudo.log
```

☐ Test and verify logging

bash

```
sudo ls
```

```
sudo cat /var/log/sudo.log
```

```
# Should show your sudo command
```

Test sudo Across All Devices

☐ Configure on Ubuntu laptop

bash

```
sudo visudo
# Add your rules
```

☐ Configure on Fedora server

bash

```
sudo visudo
# Add your rules
```

☐ Configure on Raspberry Pi devices

bash

```
# SSH into each Pi
sudo visudo
# Add your rules
```



FLASHCARD COMMANDS

Create flashcards for these commands:

1. `sudo`
2. `visudo`
3. `su`
4. `su -` (vs regular su)
5. `id`
6. `groups`
7. `whoami`
8. `sudo -l`

Important Concepts:

- Difference between `su` and `sudo`
 - Why use `visudo` instead of vim/nano
 - What `ALL=(ALL:ALL) ALL` means
 - Security implications of NOPASSWD
-

EXAM OBJECTIVES

CompTIA Linux+ (XK0-005) Exam Objectives: [https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-\(3-0\)](https://partners.comptia.org/docs/default-source/resources/comptia-linux-xk0-005-exam-objectives-(3-0))

Relevant Sections:

- 2.5 Given a scenario, manage users and groups
 - 3.2 Given a scenario, implement identity management
 - 3.3 Given a scenario, implement and configure Linux firewalls
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NOTES & REFLECTIONS

Your sudo policy decisions:

Tier 1 (Laptop/Desktop):

Tier 2 (Fedora Server):

Tier 3 (Bastion Pi):

Tier 4 (Automation Pi):

Common sudo errors encountered:

Security concerns addressed:

Completed: ☐

QUICK REFERENCE

sudoers File Format:

```
user host=(runas) commands
%group host=(runas) commands
```

Common Defaults:

```
Defaults env_reset           # Reset environment
Defaults mail_badpass       # Email on bad password
Defaults secure_path="/usr/local/bin:/usr/bin" # Safe PATH
Defaults timestamp_timeout=15 # Remember password for 15 min
```

Best Practices:

-  Always use `visudo` to edit
-  Test with `sudo -l` before logging out
-  Use groups instead of individual users when possible
-  Enable logging
-  Avoid NOPASSWD unless absolutely necessary
-  Don't give `ALL=(ALL:ALL) ALL` to service accounts

Troubleshooting:

```
bash

# Check sudo errors
sudo -l

# Verify sudoers syntax
sudo visudo -c

# View sudo attempts
sudo journalctl | grep sudo
```

Previous Week: User Management Basics

Next Week: SSH Key Authentication